

Assignment :

Convert the following statements to predicate logic.

All human that are not impoverished and intelligent are happy. Those human that read are not stupid. Jason can read and wealthy. Happy human have exciting life. Can anyone found with an exciting life.

Human(x)
Impoverished(x)
Intelligent(x)
Happy(x)
Can read(x)
Stupid(x)

Wealthy(x)
exciting life(x)

The hypotheses are :

$$\forall x (Human(x) \wedge \neg Impoverished(x) \wedge Intelligent(x) \rightarrow Happy(x))$$

$$\forall x (Human(x) \wedge read(x) \rightarrow \neg Stupid(x))$$

$$Human(Jason) \wedge read(Jason) \wedge Wealthy(x)$$

$$\forall x (Human(x) \wedge Happy(x) \rightarrow exciting\ life(x))$$

$$\exists x (Human(x) \wedge exciting\ life(x)) ?$$

$$\left. \begin{array}{l} \neg impoverished \rightarrow Wealthy(x) \\ \neg stupid \rightarrow Intelligent \end{array} \right\} \text{given human}$$

Proof

$$1) \forall x (Human(x) \wedge \neg Improvised(x) \wedge Intelligence(x) \rightarrow Happy(x))$$

$$2) \neg Improvised \rightarrow Healthy(x)$$

$$3) \forall x (Human(x) \wedge Healthy(x) \wedge Intelligence(x) \rightarrow Happy(x)) \text{ on } ①, ②$$

$$4) \forall x (Human(x) \wedge read(x) \rightarrow \neg stupid(x))$$

$$5) \neg Stupid(x) \rightarrow Intelligence(x)$$

$$6) \forall x (Human(x) \wedge read(x) \rightarrow Intelligence(x)) \text{ on } ④, ⑤$$

$$7) Human(Jason) \wedge read(Jason) \wedge Healthy(Jason)$$

$$8) Human(Jason) \quad \text{Simplification}$$

$$9) read(Jason) \quad \text{Simplification}$$

$$10) Healthy(Jason) \quad \text{Simplification}$$

$$11) Intelligence(Jason) \text{ on } ⑧, ⑩, ⑥$$

$$12) Happy(Jason) \text{ on } ⑥, ⑩, ⑪, ⑦$$

$$13) Human(Jason) \wedge Happy(Jason) \text{ on } ⑧, ⑫$$

$$14) \exists x (Human(x) \wedge Happy(x)) \text{ on } ⑬$$